Bonus Guide 5 of 5

THE POTTERY TEMPLATE VAULT

Firing Guide

Bisque & glaze firing temperatures, schedules & common errors

Firing Guide

> Firing transforms clay into ceramic. Understanding what happens inside the kiln at each temperature range — and why — gives you control over the outcome and helps you fix problems when they arise.

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1 · Kiln Types

Electric kiln (most common for studio and home potters)

- Heated by electric elements; fires in **oxidation** atmosphere (oxygen-rich).
- Predictable, programmable, easy to use.
- Produces bright, clear colors (especially blues and reds from commercial glazes).
- Standard range: Cone 04 (bisque) to Cone 10 (high-fire stoneware).

Gas kiln

- Can fire in oxidation or **reduction** atmosphere (fuel-rich, oxygen-starved).
- Reduction draws oxygen from metal oxides in the glaze, transforming colors dramatically (copper → copper red, iron → celadon green).
- Requires more skill to operate consistently.
- Produces the characteristic "flame marks" and atmospheric variation valued in wood-fire and soda-fire aesthetics.

2 · The Cone System

Pyrometric cones measure **heat work** — the combination of temperature and time. A cone with a higher number requires more heat work to melt. The system runs from **Cone 022** (very low fire) to **Cone 14** (very high fire).

Cone	Approx. temp. (°C)	Clay type	Common use
Cone 06–04	999–1063 °C	Earthenware, bisque	Bisque firing; low-fire earthenware glazes
Cone 6	1222 °C	Mid-fire stoneware, porcelain	Mid-fire glazing (most commercial studio glazes)
Cone 10	1305 °C	High-fire stoneware, porcelain	High-fire glazing; reduction firings

> ⭐ **Important**
>
> Your clay body tells you what cone it is rated to. Your glazes must be formulated for the same cone range. Firing a cone 6 glaze to cone 10 will cause it to run; firing a cone 10 clay to cone 6 will leave it porous and weak.

Witness cones

Place a witness cone (the same cone number as your target) inside the kiln, visible through the peephole. When the cone bends to 90°, the firing is complete — regardless of what the kiln controller's temperature readout says. Thermocouples drift and can mislead; witness cones are the truth.

3 · What Happens to Clay During Firing

Temperature range	What happens	Why it matters
20–120 °C	Physical water evaporates	Must be slow if any moisture remains — steam causes explosions
200–400 °C	Organic matter burns out	Carbon and organic inclusions leave; causes some porosity
573 °C	Silica inversion — quartz crystals change form	Volume change; must slow down through this range, especially on cooling
600–900 °C	Chemically combined water leaves clay minerals; clay becomes ceramic	Irreversible change; the piece can no longer be re-wet and reworked
900–1000 °C	Bisque temperature — clay is firm, porous, ready for glazing	High enough to be strong; porous enough to absorb glaze
1000–1300 °C	Flux in the clay begins to melt; vitrification begins	Clay becomes denser and less permeable; glazes melt and mature

4 · Bisque Firing

Bisque (or biscuit) firing hardens the clay piece and burns out organic matter, making it ready for glazing. The clay remains porous so it can absorb glaze.

Typical bisque schedule (Cone 04, 1000 °C)

Phase	Rate	Target temp.	Reason
Preheat	60 °C/hr	120 °C	Dry out any remaining moisture without steam build-up
Early ramp	100 °C/hr	600 °C	Burn out organic matter; pass silica inversion slowly
Fast ramp	150 °C/hr	1000 °C	Efficient heating
Hold	—	1000 °C · 15 min	Even heat throughout the kiln
Cool	Natural (kiln off)	—	Do not open before 120 °C; thermal shock cracks pieces

> ⚠️ **Warning**
> Never put greenware that feels cold and damp to the touch into a kiln. Even a slightly damp piece can generate enough steam pressure at 100 °C to crack or explode. Leave pieces in a warm room for at least 24–48 hours before bisque firing.

5 · Glaze Firing

Glaze firing melts the glaze onto the bisqueware at a higher temperature than bisque. The clay continues to mature (vitrify) during this firing.

Typical glaze firing schedule (Cone 6, 1222 °C)

Phase	Rate	Target temp.	Reason
Ramp 1	100 °C/hr	600 °C	Gradual heat-up; pass silica inversion slowly
Ramp 2	150 °C/hr	1000 °C	Burn off remaining carbon in glazes
Ramp 3	100 °C/hr	1222 °C	Slow final approach — even maturation, reduce running risk
Hold	—	1222 °C · 10 min	Even heat throughout; smooth out glaze surface
Crash cool	150 °C/hr	1100 °C	Fast cool from peak helps glaze surface quality and color
Natural cool	Natural	—	Must slow through 573 °C (silica inversion)

> ⭐ **Important**
>
> Do not open the kiln until it reaches below 100 °C. The thermal shock of cool air on a hot piece can cause crazing (tiny cracks in the glaze) or worse, can crack the piece entirely.

6 · Kiln Loading

Bisque loading

- Pieces can touch each other and be stacked (because there is no glaze to stick them together).
- Heavy pieces go on lower shelves, lighter on top.
- Stack plates rim-to-rim with wadding between.
- Leave some space between pieces for air circulation.

Glaze loading

- **No glazed surfaces may touch** — they will fuse together permanently.
- Leave at least 5–10 mm between glazed pieces.
- The foot of every piece must be clean of glaze (wax or wipe clean).
- Use kiln wash (a refractory coating) on shelves to protect them from dripping glaze.
- Place a catch tray under pieces with fluid or heavy glazes.

> 💡 **Tip**
>
> Load from the back of the kiln to the front, and from the bottom shelf up. This way you never have to reach over loaded pieces.

7 · Firing Schedules

Your kiln controller accepts a firing schedule — a sequence of ramp rates (°C per hour) and target temperatures with optional holds. Three schedules cover most studio needs:

Schedule	Use	Key feature
Bisque (slow)	Thick or wet pieces; first firings	60 °C/hr preheat to 120 °C; slow through 200–600 °C
Bisque (fast)	Thin, thoroughly dry pieces	Start at 100 °C/hr; no hold below 600 °C
Glaze (cone 6 standard)	Mid-fire stoneware and porcelain	Slow final ramp 100 °C/hr; 10-min hold at peak; crash-cool to 1100 °C

8 · Reading a Kiln Firing

After bisque

- Pieces should be uniformly light in color, hard and ring when tapped (like a ceramic, not a dull thud).
- Black spots or grey areas indicate incomplete carbon burnout — increase preheat time or slow down through 400–700 °C next firing.

After glaze firing

- Witness cone should be bent to 90°. If only half-bent, the kiln under-fired — refire.
- Check for crawling, pinholes and runs — note them for adjustment.
- Any glaze that has stuck to shelves must be ground off (corundum stone) before refiring that shelf.

9 · Common Firing Problems and Solutions

Problem	Likely cause	Solution
Piece explodes in the kiln	Moisture trapped in clay; air pocket sealed during building	Ensure thorough drying; slow preheat ramp; score and compress walls during building to close any air pockets
Pieces crack during cooling	Opened kiln too soon; rapid cool through 573 °C	Do not open until <120 °C; ensure slow cool through 573 °C
Glaze runs and sticks to shelf	Glaze too thick or too fluid; firing over-temperature	Thin glaze; check witness cone vs. controller; use kiln wash
Under-fired glaze (dull, dry)	Controller thermocouple drifted low; insufficient hold	Use witness cones; add 10-min hold at peak; recalibrate controller
Carbon spotting (black marks)	Carbon burnout incomplete; organic matter in clay	Slow the 400–700 °C range; leave kiln lid/bung slightly open below 600 °C for ventilation
Crazing after glaze firing	Glaze-clay thermal expansion mismatch; too-rapid cooling	Switch to a glaze matched to your clay; slow the cooling through 573 °C; try a controlled anneal hold at 550 °C
Pieces stuck together	Glazed surfaces touching; glaze on foot	Leave adequate space between pieces; always clean feet; use kiln wash

Pre-firing checklist

- Are all pieces thoroughly dry before loading for bisque?
- Are glazed pieces not touching each other or the kiln walls?
- Are all foot rings clean of glaze?
- Is there kiln wash on the shelves?
- Is a witness cone placed where I can check it through the peephole?
- Is the firing schedule set correctly in the controller?

You now have all five bonus guides. Combined with your 100 templates, you have everything you need to design, build, decorate and fire your pottery with confidence. Enjoy the process.